

Samia Rahman

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Education

University of Alberta

BSc Specialization in Computing Science

Edmonton, Alberta

Sept. 2022 – May 2027

Experience

Data Science Intern

May 2026 – Aug 2026

Capital Power

Edmonton, AB

- Built an end-to-end MLOps pipeline for turbine failure prediction at a 1.6GW energy facility using Snowflake, Databricks, and Azure DevOps to identify equipment-level failure drivers for root-cause analysis.
- Benchmarked PyOD anomaly detection models (Isolation Forest and others) against statistical baselines for early fault detection.

Software Engineer Intern

May 2025 – Dec 2025

City of Edmonton – Financial & Corporate Services

Edmonton, AB

- Automated processing of millions of records for 500+ users by designing and testing scalable backend data pipelines integrating 20 SAP Ariba REST APIs with Python (Pandas, NumPy, SQLAlchemy) and SQL Server.
- Reduced reporting discrepancies by 30% across supply chain systems by implementing automated validation, monitoring, and data quality controls.

Machine Learning Engineer Intern

Jan 2025 – Apr 2025

Process Data Analytics and Smart Automation Lab (PDASA)

Edmonton, AB

- Reduced manual bitumen extraction monitoring by **80%** by deploying a PyTorch A3C reinforcement learning agent with CNN-LSTM encoders for closed-loop valve regulation via camera sensors.
- Cut training time by **35%** by migrating a legacy RL system from TensorFlow 1.x to PyTorch and redesigning training pipelines.

ML & Robotics Research Assistant

Sep 2024 – Present

University of Alberta

Edmonton, AB

- Improved reinforcement learning efficiency by **50%** on robotic control tasks by building State-Conditioned Linear (SCL) action maps with VAEs, reducing control complexity from 6D to 3D.
- Achieved **80x** lower reconstruction error by training VAEs across 6 latent dimensions to identify the optimal action representation for robot control.

Projects

InclusiTrip: Accessible Travel Matching Platform | FastAPI, Python, HTML, CSS, JavaScript

May 2026

- Improved accessibility-aware travel discovery by building a FastAPI platform that matches travelers to services based on mobility, sensory, cognitive, neurodivergent, and temporary accessibility needs.

UNTAPPED: Artist Management Web App | React, Node.js, PostgreSQL, OAuth2, AWS S3

April 2026

- Built and deployed a production multi-tenant portal for ACPA Foundation serving 50+ artists, eliminating manual admin workflows through invoice approval pipelines, role-based access control, and PIPEDA-compliant AWS S3 document storage.

IoT Smart Home Test Automation | Java, JUnit 5, Docker, GitHub Actions

Dec 2025

- Achieved 80%+ branch coverage on state consistency logic by designing a TDD test suite (black-box, boundary value analysis) and automating it with a Docker + GitHub Actions CI/CD pipeline running on every commit.

Technical Skills

Full-Stack Development: Python, Java, C, TypeScript, JavaScript, SQL, React, Node.js, Express, HTML/CSS, REST APIs, PostgreSQL, MongoDB

AI/ML: Machine Learning, Reinforcement Learning, Large Language Models (LLMs), Generative AI, Prompt Engineering, RAG, Embeddings, PyTorch, TensorFlow

Data Engineering & Analytics: ETL Pipelines, Pandas, NumPy, Power BI, Tableau, Jupyter Notebook

Cloud & DevOps: AWS, Docker, Firebase, Linux, Git, GitHub Actions, CI/CD, Snowflake, Databricks

Leadership

Ada's Team

May 2025 – present

VP Finance – Conferences

University of Alberta

- Managed \$45K+ budget and end-to-end conference logistics for Ada's Team, a diversity-in-tech club, securing industry sponsorships to fund travel and programming for 200+ underrepresented CS students across Edmonton